

SAURABH KATARIA, Ph.D.

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RESEARCH INTERESTS

Robust Signal processing, Audio and Language Processing, Paralinguistics, Foundation models

EXPERIENCE

External Research Collaborator, *School of Nursing, Emory University* Apr'26 - present
LLMs for Dementia, Emotion Recognition using Audio Foundation Models
Mentor: Prof. Xiao Hu

Postdoctoral Fellow, *Center for Data Science, School of Nursing, Emory University, USA* Dec'23 - Mar'26
Photoplethysmography-sensor-based foundation models for human health assessment Atlanta, GA, USA
Mentor: Prof. Xiao Hu
Affil.: *Children's Healthcare of Atlanta (CHOA) Hospital, Georgia*; Collab.: *Nihon Kohden Digital Health Solutions, California*

EDUCATION

Doctor of Philosophy (Ph.D.), Electrical Engineering, *Johns Hopkins University, USA* Aug'17 - Jul'23
Thesis: "Robust Speaker Recognition using Perceptual and Adversarial Speech Enhancement" [\[link\]](#) GPA: 3.8/4
Advisors: Prof. Jesús Villalba, Prof. Najim Dehak
Affil.: *Center for Speech and Language Processing (CLSP), Human Language Technology Center of Excellence (HLTCOE)*

Master of Science in Engineering (M.S.E.), *Johns Hopkins University, USA* Aug'17 - May'20
Electrical and Computer Engineering GPA: 3.8/4

B.Tech. - M.Tech. dual degree program, *Indian Institute of Technology (IIT) Kanpur, India* Jul'12 - Jun'17
• **Master of Technology (M.Tech.)**, Electrical Engineering GPA: 9.0/10
Thesis: "Unsupervised and Supervised Saliency Estimation Methods for Affective Speech"
Advisors: Prof. Tanaya Guha, Prof. Rajesh Hegde
• **Bachelor of Technology (B.Tech.)**, Electrical Engineering GPA: 8.4/10
• **Minor in Artificial Intelligence**, Computer Science and Engineering

AISSCE (XII), CBSE Board, DAV Sr. Sec. Public School, Barmana, H.P. 95.0%

AISSE (X), CBSE Board, DAV Sr. Sec. Public School, Barmana, H.P. 9.6/10

INTERNSHIPS

NLP Research Intern, *Tencent America* May'20 - Aug'20
Mentors: Dr. Shi-Xiong (Austin) Zhang, Dr. Dong Yu Seattle, WA, USA (remote)
Multi-channel Audio-Visual speech enhancement and speaker recognition [\[link\]](#)

Student Researcher, *Frederick Jelinek Memorial Summer Workshop (JSALT)* [\[link\]](#) May'19 - Aug'19
Venue: École de Technologie Supérieure (ÉTS), Organizer: CLSP, Johns Hopkins University Montreal, Canada
Mentors: Prof. Jesus Villalba, Prof. Najim Dehak, Leader: Dr. Paola Garcia
Speech enhancement using auxiliary DNNs [\[link\]](#)

Research Intern, *INRIA* May'16 - Aug'16
Mentor: Dr. Antoine Deleforge, Leader: Dr. Rémi Gribonval (Team PANAMA) Rennes, France
Estimating room acoustics and data simulation [\[link\]](#)

Research Intern, Tandon School of Engineering, *New York University* May'15 - Aug'15
Mentor: Prof. Siddharth Garg New York City, NY, USA
De-anonymizing social network graphs

NEWS ARTICLES

• SIAM - <https://www.siam.org/publications/siam-news/articles/foundation-models-for-physiological-data/>

AWARDS

- Emory University Postdoctoral Association Professional Development Travel Award 2025 (research merit-focused)
- Johns Hopkins University ECE department graduate PhD fellowship (2017-2018)
- Best project award in the course EE698M (Topics in Image and Signal Processing; Prof. Tanaya Guha), 2016 titled “Scene intensity estimation and ranking for movie scenes through direct content analysis”
- Awarded three travel grants by Dept. of Science and Technology (DST) India, IIT Kanpur, and EE dept resp. (2015)

JOURNALS

1. Fattahi D., Yan R., **Kataria S.**, Chen Z., and Hu X., *A Comprehensive Inference-Time Augmentation Framework in Physiological Signals: Application to PPG-Based AF Detection*, 2026, *Physiological Measurement* [pdf]
2. Chen, Z., Ding, C., **Kataria, S.**, Yan, R., Wang, M., Lee, R. and Hu, X., 2025. *GPT-PPG: a GPT-based foundation model for photoplethysmography signals*. *Physiological Measurement*, 46(5), p.055004. [pdf]
3. **Kataria, S.**, Villalba, J., Moro-Velázquez, L., Żelasko, P. and Dehak, N., *Time-domain speech super-resolution with GAN based modeling for telephony speaker verification*. *IEEE/ACM Transactions on Speech, Audio, and Language journal* 2024 [pdf]

CONFERENCE PAPERS

1. Chen Z., **Kataria S.**, Wang M., Hu X., *Bridging ECG and PPG: Latent-Space Prediction for Robust Physiological Analysis*, to appear in KDD 2026 (**CORE A* CS conference**)
2. Panchumarthi, L. Y., Meng, Z., Lu, A., Kim, T., **Kataria, S.**, Fedorov, A., Bold, D., Gow, B., Pollard, T. J., Zègre-Hemsey, J., Dzikowicz, D., Hu, X., van Assen, M., Wright, D. W., Kumar, L., Xiao, R. *CardioFM: A multimodal foundation model for ECG representation learning using waveform reconstruction and clinical text alignment*. 2026 International Society for Computerized Electrocardiology (ISCE) Conference.
3. Ni, J., **Kataria, S.**, Tang, S., Yang, C., Hu, X. and Jin, W., 2025. *Ppg-distill: Efficient photoplethysmography signals analysis via foundation model distillation*. *NeurIPS 2025 TS4H workshop* (**Spotlight**) [pdf]
4. Wang, M., Yan, R., Li, C., **Kataria, S.**, Hu, X., Clark, M., Ruchti, T., Buchman, T.G., Bhavani, S.V. and Lee, R.J., 2025. *Estimating Clinical Lab Test Result Trajectories from PPG using Physiological Foundation Model and Patient-Aware State Space Model—a UNIPHY+ Approach*. *NeurIPS 2025 TS4H workshop* [pdf]
5. **Kataria, S.**, Xiao, R., Ruchti, T., Clark, M., Lu, J., Lee, R.J., Grunwell, J. and Hu, X., 2025, July. *Continuous cardiac arrest prediction in icu using ppg foundation model*. In 2025 47th Annual International Conference of the IEEE Engineering in Medicine and Biology Society (EMBC) (pp. 1-6). IEEE. [pdf] (**Oral**)
6. Meng, Z., Panchumarthi, L.Y., **Kataria, S.**, Fedorov, A., Zègre-Hemsey, J., Hu, X. and Xiao, R., 2025. *Fusion of ECG foundation model embeddings to improve early detection of acute coronary syndromes*. *MEDINFO 2025*
7. **Kataria, S.**, Villalba, J., Moro-Velázquez, L., Thebaud, T. and Dehak, N., 2023. *Self-FiLM: Conditioning GANs with self-supervised representations for bandwidth extension based speaker recognition* *INTERSPEECH 2023* [pdf] (**Oral**)
8. Villalba, J., Borgstrom, J., Jahan, M., **Kataria, S.**, Garcia-Perera, L.P., Torres-Carrasquillo, P.A. and Dehak, N., 2023. *Advances in Language Recognition in Low Resource African Languages: The JHU-MIT Submission for NIST LRE22*. (*INTERSPEECH 2023*) [pdf] (**Oral**)
9. **Kataria, S.**, Villalba, J., Moro-Velázquez, L. and Dehak, N., *Joint domain adaptation and speech bandwidth extension using time-domain GANs for speaker verification*. *INTERSPEECH 2022* [pdf] (**Oral**)
10. Joshi, S.*, **Kataria, S.***, Villalba, J. and Dehak, N., *AdvEst: Adversarial Perturbation Estimation to Classify and Detect Adversarial Attacks against Speaker Identification*. *INTERSPEECH 2022* [pdf]
11. Shao, Y., Villalba, J., Joshi, **S.**, **Kataria, S.**, Dehak, N. and Khudanpur, S., *Chunking Defense for Adversarial Attacks on ASR*. *INTERSPEECH 2022* [pdf]
12. Joshi, S., **Kataria, S.**, Shao, Y., Żelasko, P., Villalba, J., Khudanpur, S. and Dehak, N., *Defense against Adversarial Attacks on Hybrid Speech Recognition using Joint Adversarial Fine-tuning with Denoiser*. *INTERSPEECH 2022* [pdf]
13. Villalba, J., Borgstrom, B.J., **Kataria, S.**, Rybicka, M., Castillo, C.D., Cho, J., Garcia-Perera, L.P., Torres-Carrasquillo, P.A. and Dehak, N., *Advances in Cross-Lingual and Cross-Source Audio-Visual Speaker Recognition: The JHU-MIT System for NIST SRE21*. *Odysey 2022*. [pdf]

14. Villalba, J., Borgstrom, B.J., **Kataria, S.**, Cho, J., Torres-Carrasquillo, P.A. and Dehak, N., *Advances in Speaker Recognition for Multilingual Conversational Telephone Speech: The JHU-MIT System for NIST SRE20 CTS Challenge*. Odyssey workshop 2022 [\[pdf\]](#)
15. **Kataria, S.**, Villalba, J. and Dehak, N., *Perceptual loss based speech denoising with an ensemble of audio pattern recognition and self-supervised models*. ICASSP 2021 [\[pdf\]](#)
16. **Kataria, S.**, Villalba, J., Żelasko, P., Moro-Velázquez, L. and Dehak, N., *Deep feature cyclegans: Speaker identity preserving non-parallel microphone-telephone domain adaptation for speaker verification*. INTERSPEECH 2021 [\[pdf\]](#)
17. **Kataria, S.***, Zhang, S.X.* and Yu, D., *Multi-Channel Speaker Verification for Single and Multi-Talker Speech*. INTERSPEECH 2021 [\[pdf\]](#)
18. **Kataria, S.**, Nidadavolu, P.S., Villalba, J., Chen, N., Garcia-Perera, P. and Dehak, N., *Feature enhancement with deep feature losses for speaker verification*. ICASSP 2020 [\[pdf\]](#)
19. Nidadavolu, P.S., **Kataria, S.**, Villalba, J., Garcia-Perera, P. and Dehak, N., *Unsupervised feature enhancement for speaker verification*. ICASSP 2020 [\[pdf\]](#)
20. **Kataria, S.**, Nidadavolu, P.S., Villalba, J. and Dehak, N., *Analysis of deep feature loss based enhancement for speaker verification*. Odyssey 2020 [\[pdf\]](#)
21. Villalba, J., Garcia-Romero, D., Chen, N., Sell, G., Borgstrom, J., McCree, A., Garcia-Perera, L.P., **Kataria, S.**, Nidadavolu, P.S., Torres-Carrasquillo, P.A. and Dehak, N., *Advances in speaker recognition for telephone and audio-visual data: the JHU-MIT submission for NIST SRE19*. Odyssey 2020 [\[pdf\]](#)
22. García, P., Villalba, J., Bredin, H., Du, J., Castan, D., Cristia, A., Bullock, L., Guo, L., Okabe, K., Nidadavolu, P.S. and **Kataria, S.**, et al., *Speaker detection in the wild: Lessons learned from JSALT 2019*. Odyssey 2020 [\[pdf\]](#)
23. Nidadavolu, P.S., **Kataria, S.**, Villalba, J. and Dehak, N., *Low-resource domain adaptation for speaker recognition using cycle-gans*. ASRU 2019 [\[pdf\]](#)
24. **Kataria, S.**, Gaultier, C. and Deleforge, A., *Hearing in a shoe-box: binaural source position and wall absorption estimation using virtually supervised learning*. ICASSP 2017 [\[pdf\]](#) (Oral)
25. Gaultier, C., **Kataria, S.** and Deleforge, A., *VAST: The virtual acoustic space traveler dataset*. In International Conference on Latent Variable Analysis and Signal Separation (ICA-LVA 2015) [\[pdf\]](#) (Released dataset)
26. Parthasarathy, A., **Kataria, S.**, Kumar, L. and Hegde, R.M., *Representation and modeling of spherical harmonics manifold for source localization*. ICASSP 2015 [\[pdf\]](#) (Oral)

OTHERS (ABSTRACT, SYMPOSIUM, ETC.)

1. Zhang, Z., Song, M., Gupta, P., Yan, R., Wang, C.-Y., **Kataria, S.**, Dunlap, P., Hu, X., Michalowski, M., & Topaz, M. (2026). Evaluating wearable camera systems for multimodal AI in home healthcare: A feasibility study. AMIA Annual Symposium.
2. Brown, S.R., Xie, Y., Grunwell, J., Hu, X., Yang, C., **Kataria, S.**, Bold, D., and Lu, J., *Early Prediction of Pediatric In-Hospital Cardiac Arrest Using a Tabular Foundation Model* (Critical Care Congress by Society of Critical Care Medicine (SCCM'25)) [\[link\]](#)
3. Brown, S.R., Xie, Y., Grunwell, J., Hu, X., Yang, C., **Kataria, S.**, Bold, D., and Lu, J., *External Validation of a Model to Predict Cardiac Arrest in the Pediatric Intensive Care Unit*, (Pediatric Research for the Future of Children conference 2026)
4. **Kataria, S.**, Ermis, A., Panchumarthi, L.Y., Wang, M. and Hu, X., 2025. *Vision4PPG: Emergent PPG Analysis Capability of Vision Foundation Models for Vital Signs like Blood Pressure*. (IEEE BHI'25) [\[pdf\]](#)
5. **Kataria, S.**, Lu, J., Xiao, R., Ruchti, T., Clark, M., Bergmann, J. and Hu, X., 2025. Cardiac arrest classification using large PPG foundation model representations. (International Conference on Complex Acute Illness (ICCAI'24)) [\[pdf\]](#)

MANUSCRIPTS UNDER CONSIDERATION

1. **Kataria, S.**, Wu, Y., Chen, Z., Kwak, H.G., Xu, Y., Panchumarthi, L.Y., Xiao, R., Lu, J., Ermis, A., Zhao, A. and Yan, R., *Generalist vs Specialist Time Series Foundation Models: Investigating Potential Emergent Behaviors in Assessing Human Health Using PPG Signals*. In submission to *Nature Communications*. [\[pdf\]](#) (JOURNAL)

2. **Kataria, S.**, Fattahi, D., Wang, M., Xiao, R., Clark, M., Ruchti, T., Mai, M. and Hu, X., *Wav2Arrest 2.0: Long-Horizon Cardiac Arrest Prediction with Time-to-Event Modeling, Identity-Invariance, and Pseudo-Lab Alignment*. In submission to *Biomedical Signal Processing and Control*. [\[pdf\]](#) (**JOURNAL**)
3. **Kataria, S.** and Hu, X., *Prompt Amplification and Zero-Shot Late Fusion in Audio-Language Models for Speech Emotion Recognition*. In submission to *Computer Speech and Language*. [\[pdf\]](#) (**JOURNAL**)
4. Panchumarthi, L. Y., **Kataria, S.**, Wu, Y., Hu, X., Fedorov, A., Kwak, H., *Context-Dependent Age-Group performance hierarchies limit fairness interventions in PPG-based heart rate prediction*, In submission to *PLOS Digital Health* [\[pdf\]](#) (**JOURNAL**)
5. Wang, M., Yoshida, A., Xu, Y., Shen, X., Yan, R., Fattahi, D., Li, C., Hossain, M. S. B., Chen, Z., **Kataria, S.**, Xiao, R., Clark, M., Ruchti, T., Brown, S., Mai, M., Bhavani, S. V., Wachi, A., Gros, S., Liu, X., Matono, A., Kim, K., Yang, C., Fujisawa, K., Hu, X., *A long-context, personalized framework linking second-scale physiological signals to day-scale clinical trajectories*. In submission to *Nature Digital Medicine*. (**JOURNAL**)
6. Lu, A., Castle, L., Cho, S., Meng, Z., Yan, R., **Kataria, S.**, Westerman, S., Hu, X., Xiao, R., *ECG foundation model features improve atrial fibrillation alarm precision in intensive care monitoring*, IEEE International Workshop on Machine Learning for Signal Processing (MLSP) 2026 (**CONFERENCE**)

ORGANIZER

1. (**LEAD**) Symposium w/ University of Bologna | IEEE EMBC'25 | Foundation models and Pain management in Oncology
2. 8th Mid-Atlantic Student Colloquium on Speech, Language and Learning (MASC-SLL 2020), Program Organizing Committee, University of Maryland, College Park

MISC

- Member, ACM (Association for Computing Machinery)
- Full invited member, Sigma Xi: The Scientific Research Honor Society, 2025
- Expert peer reviewer for *BMJ Digital Health* & *AI* 25-26
- New Graduate Student Welcome Committee member at Johns Hopkins University

CHALLENGES/EVALUATIONS

(w/ MIT Lincoln Lab) *NIST Language Recognition Evaluation: '22, '17* • *NIST Speaker Recognition Evaluation: '21-'18*
 DARPA GARD (Guaranteeing AI Robustness Against Deception) • DARPA RED (Reverse Engineering of Deceptions)

TALKS

1. Faculty Seminar, IIT Jodhpur, Jun'26
2. “*Generalist vs Specialist Foundation Models for Human Health Assmt.*”, Bridge2AI Conf., NIH Neuroscience Ctr, May'25
3. “*Robust Speaker Verification using speech enhancement*”, Brookhaven National Lab, NY, May'23
4. “*CycleGAN variants for Speech Domain Adaptation and Bandwidth Expansion*”, CLSP seminar, JHU, Nov'21

MENTEES

Dr. Sonal Sunil Joshi (Junior PhD)

TEACHING/COURSE ASSISTANCE

1. Machine Learning for Medical Applications (2021, EN.520.639, Johns Hopkins University)
2. Computational Modeling for Electrical and Computer Engineering (2020, EN.520.123, Johns Hopkins University)
3. Machine Learning for Signal Processing (2020, EN.520.612, Johns Hopkins University)
4. Modeling and Representation Techniques for Images (2017, EE698K, IIT Kanpur)
5. Image processing (2016, EE604A, IIT Kanpur)

REVIEWING SERVICE

IEEE SPCOM'26, Engineering Research Express ('25), JMIR Medical Informatics journal ('25), ICASSP ('26, '25, '23), Engineering Applications of Artificial Intelligence (EAAI'25), Sensors journal ('25), TMLR (multiple), Multimedia Systems (Journal), BHI'25, The Journal of Supercomputing, TDSC (IEEE Transactions on Dependable and Secure Computing), TASLP (IEEE/ACM Transactions on Audio, Speech, and Language Processing) (multiple), INTERSPEECH ('26-'23), PMEA journal (multiple), IJCNN'25, BMVC'24, ICLR ('23, '21), CISS ('23), ACML ('22), Speech Communication (multiple), NeurIPS'20